

T M V S G Pavan

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EXPERIENCE

XFactr Softwares Private Limited (Xfactr.ai) | July 2026 – Present

AI & ML Engineer – Distributed Systems & AI Platforms · Bangalore

- Build a C++17 tenant agent-package import runtime and deployment orchestration with gRPC/Protobuf, desired-state reconciliation, idempotency, leases, and fencing.
- Develop a Go edge/relay gateway with sticky sessions and streaming; implement durable continuation using acknowledged sequences and replay without re-execution, plus cancellation and recovery.
- Implement OAuth 2.0/OIDC, PKCE, JWT/JWKS, RBAC, tenant isolation, and human approvals; secure workload admission with signatures, mTLS, protected Unix sockets, and checkpoint/restore.
- Build typed React/TypeScript workspaces for tenant administrators, managers, employees, and operators, backed by shared API contracts.
- Develop AgentHRMS / DeepSeek Harness tool-calling workflows for leave, attendance, payroll, and administration using Python, Go, React, FastAPI, and WebSockets; support durable conversations, write approvals, artifacts, and interruptible LiveKit/Deepgram voice interactions.
- Validate changes through Go race tests, C++ Meson/Ninja builds, React type checks, Playwright, Docker gates, and negative identity and replay tests.

Netskope | Jan 2026 – June 2026

AI Intern – Intrusion Prevention System (IPS) Team · Bangalore

- Built LangChain tool-calling agents for VirusTotal reputation, WHOIS, User-Agent anomalies, and process analysis; achieved 80% manual analyst workload reduction, exceeding a 50% target.
- Processed 4.25M IPS events into 72 features using label conflict resolution, stratified sampling, and TF-IDF + Truncated SVD across URL, User-Agent, process, and hostname fields.
- Trained LightGBM and LSTM models with Optuna tuning and MLflow tracking; deployed a Docker-containerized REST service with monitoring, drift detection, and retraining.

BITS Pilani Digital | June 2025 – July 2026

Content Developer & Teaching Assistant – Data & Machine Learning · M.Sc. programmes

- Developed course content for Data Preprocessing, Machine Learning, Unsupervised Learning, and selected Data Mining topics, including presentation decks, written learning assets, quizzes, discussion prompts, and practice-lab exercises.
- Served as Teaching Assistant for Data Preprocessing across multiple trimesters, supporting 500+ students per trimester through technical query resolution, assessment administration, answer-script correction, feedback, and consistent evaluation.

EDUCATION

BITS Pilani, Goa Campus | 2024 – 2026

M.E. Computer Science – CGPA: 9.67 · Goa

Sri Venkateswara University College of Engineering | 2020 – 2024

B.Tech Computer Science & Engineering – CGPA: 8.6 · Tirupati

SELECTED PROJECTS & RESEARCH

BITS Social – Full-Stack Social Platform | Apr 2025 – Present

- Built a React 19 frontend and Express 5 / Prisma backend with PostgreSQL and a three-node CockroachDB cluster; packaged six services in Docker Compose and deployed the frontend on Vercel.
- Implemented Kafka/Redpanda fan-out-on-write and Redis sorted-set timelines trimmed to 2,000 entries; added JWT and Google OAuth, a Redis sliding-window rate limiter, Zod validation, Pino logging, and MinIO/S3 media storage.

Phishing Detection & Explainable Deep Learning | Aug 2024 – Dec 2024

- Evaluated transformer embeddings with recurrent and feedforward classifiers; achieved 99.9% accuracy with RoBERTa-LSTM in the reported research evaluation.
- Used Captum Integrated Gradients for attribution and assessed explanation faithfulness; compared standalone and concatenated embeddings for phishing detection.

3D NoC Mapping & TSV Optimization – M.E. Thesis | Feb 2025 – Nov 2025

- Optimized IP-core mapping and TSV placement in 3D Network-on-Chip using Tabu Search and Gurobi ILP, targeting communication cost and thermal hotspots with Noxim simulation.
- Explored learning-based optimization with Active Search, MPNN, and Pointer Networks; the heuristic approach achieved 95–98% of optimal quality with 10x lower runtime in the reported evaluation.

TECHNICAL SKILLS

Languages: Go, C++17, Python, TypeScript, C, SQL

Distributed Systems & Security: gRPC, Protobuf, streaming, durable execution, idempotency, OAuth 2.0, OIDC, PKCE, JWT/JWKS, RBAC, mTLS

AI & ML: LangChain, tool-calling agents, agent workflows, DeepSeek Harness, PyTorch, scikit-learn, LightGBM, Transformers, LSTM, Captum, Optuna, MLflow

Web & Data: React, Express, FastAPI, WebSockets, Prisma, PostgreSQL, CockroachDB, Redis, Kafka/Redpanda, LiveKit, Deepgram, MinIO/S3

Engineering: Docker, Docker Compose, Git, Linux, Playwright, Meson/Ninja, CI/CD, monitoring and drift detection

PUBLICATION

- PhishBuster – EAI Broadnets 2024. Published 22 Jan 2026. DOI: 10.1007/978-3-032-14350-1_4